MILITARY SPECIFICATION

RADIO SET AN/SRC-23A

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the requirements of the radio frequency (RF) system which, through compatible operation with the parts of the antenna coupler group AN/SRA-34A(V), provides the high frequency RF functions of the Naval Tactical Data System (NTDS) "A" Link as well as shipboard general purpose communications.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, unless otherwise specified herein, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-C-17	- Cables, Radio Frequency; Coaxial, Dual Coaxial, Twin Conductor, and Twin Lead
MIL-S-901	- Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements for
MIL-Q-9858	- Quality Program Requirements
MIL-E-16400	- Electronic, Internal Communication and Navigation Equipment, Naval Ship and Shore; General Specification for
MIL-E-16910	- Interference Measurement, Electromagnetic, Methods and Limits
MIL-W-21965	- Water Cooling of Shipboard Electronic Equipment; General Specification for
MIL-C-24056	- Cabinets, Electrical Equipment CY-4516/S (Single Bay) (Naval Shipboard Use)

STANDARDS

MILITARY

MIL-STD-108	- Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-188	- Military Communication System Technical Standards

FSC 5820

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MIL-STD-480	- Configuration Control - Engineering Changes, Deviations and Waivers
MIL-STD-781	- Reliability Tests: Exponential Distribution
MIL-STD-1399	- Interface Standard for Shipboard Systems

PUBLICATIONS

MILITARY

NAVSHIPS 0967-191-5010 (Formerly 93152)	- Radio Set Control C-1138A/UR
NAVSHIPS 93820	- Handbook for the Prediction of Shipboard and Shore Electronic Equipment Reliability

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General. The AN/SRC-23A(V) shall conform to the applicable detail requirements of MIL-E-16400 for shipboard production equipment, except as otherwise specified herein.

3.1.1 Environment. The AN/SRC-23A(V) shall be designed to meet the Range 4 temperature requirements of MIL-E-16400.

3.1.2 Uniformity. The electronic assemblies of the AN/SRC-23A(V) shall be electrically and physically interchangeable with those of the Communication Central AN/SRC-16A to the maximum extent possible. Insofar as practical, the design of equipment shall employ a minimum of types and kinds of parts. Assemblies performing similar functions shall be identical in design and construction.

3.1.3 Reliability. The equipment shall conform to the reliability test requirement of 4.5.

3.1.4 Shock. The AN/SRC-23A(V) shall be capable of meeting the grade A, class I, type A shock requirements of MIL-S-901.

3.1.4.1 Locking devices. The use of locking devices on controls shall be as necessary.

3.1.5 Cooling. Water cooling for the equipment shall be as specified herein; water cooling in accordance with method 1 of MIL-W-21965 shall be provided. The maximum allowable heat dissipation to the local environment shall be 1000 watts or 25 percent of the total heat generated in each equipment cabinet, whichever is the lesser.

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3.1.6 Enclosure. The AN/SRC-23A(V) shall be splashproof in accordance with MIL-STD-108.

3.1.7 Primary power. The AN/SRC-23A(V) shall be capable of operating within the requirements of this specification when connected to a 115 volt or 115 and 440 volt, 400 Hz, 3-phase power source with characteristics conforming to type I of MIL-STD-1399, section 103.

3.1.7.1 Utilization requirements. The AN/SRC-23A(V) shall conform to the requirements specified for utilization equipment, type I of MIL-STD-761.

3.1.8 Illumination. The illumination requirements of dials, controls and lights shall be governed by conditions of equipment installation in an illuminated compartment aboard ship.

3.1.9 Cabling. All cabling, loops, lines, and trunks shall be in accordance with MIL-STD-188.

3.1.10 Acoustical noise. Acoustical noise shall be held to an absolute minimum and shall not hinder normal operator function within the immediate area.

3.1.11 Blown fuse indicators. Wherever fuses are used for protection against damaging overloads, blown fuse indicators shall be provided.

3.1.12 Connectors. Required mating cable connectors shall be furnished as part of the equipment. Stuffing tubes, if used, shall also be furnished.

3.1.13 Electromagnetic interference emission and susceptibility. The AN/SRC-23A(V) shall withstand the tests specified in MIL-I-16910.

3.2 System requirements.

3.2.1 Equipment. The AN/SRC-23A(V) shall consist of automatically tuned transmitting and receiving equipment for general purpose communications. The radio shall provide a dual single-side band (SSB) capability with both side bands available for use independently in any of the modes of operation in simplex or duplex.

3.2.2 RF emission. The transmit and receive channels shall each be capable of independent operation on frequencies in the 2.0000 to 29.9999 MHz range selectable in 100 Hz steps, with optional continuous tuning between 100 Hz steps available on the receiver.

3.2.2.1 Modes. Operation in any of the following modes shall be possible:

(a)	SSB telephony, suppressed carrier	(A3J)
(b)	Two independent sidebands, telephony	(A3B)
(c)	Composite transmissions	(A9B)
(d)	CW telegraphy	(A1)
(e)	Frequency shift telegraphy	(F1)

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(f) Compatible amplitude modulated (AM) telephony:
Transmit; SSB, full carrier (A3H)
Receive; double-sideband, (AM) (A3)
(g) Diversity phase shift keying (Data transmission) (F9)

3.2.2.2 FSK mode. The frequency-shift keying (FSK) mode of operation shall be in accordance with 4.5.1.5.2 of MIL-STD-188. Provisions shall be made for operation at nominal 850 Hz frequency shift at keying speeds from that required of 60 words-per-minute (WPM) teletypewriter operation to that required by the four channel time division multiplex AN/UGC-1 operating at a rate of 75 baud per channel.

3.2.2.2.1 Narrow band conversion. Equipment design shall allow for conversion to narrow band FSK by replacement of specific modules by similar units at a later date.

3.2.2.3 Upper sideband. The upper sideband shall be utilized for the transmission and reception of CW FSK and compatible AM signals.

3.2.2.4 IF translation. Translation of the signals from or to an intermediate frequency (IF) of 500 kHz shall be provided and shall conform to the following conditions:

(a) No radiation takes place on 500 kHz beyond 10 micromicro watts to a matched load at the antenna input terminal.
(b) The receiver IF rejection against 500 kHz is at least 100 decibels (dB) including entry of the 500 kHz signal by means of antenna terminals, case penetrations, power input leads, headphone terminals, and control or other auxiliary leads.

3.2.3 Flexibility. The AN/SRC-23A(V) shall contain sufficient flexibility to allow for variable configurations of multiple channel installations on a building block concept. Compatible operation with the Antenna Coupler Group AN/SRA-34A(V) shall be possible on the same basis. In addition, the AN/SRC-23A(V) shall be compatible on the same basis as specified in 3.2.3, where a shipboard manual RF transfer panel is used in lieu of the AN/SRA-34A(V) RF switching matrix.

3.2.4 Frequency stability. Frequency determining circuits shall be phase locked to an external frequency standard to maintain the accuracy of the standard. Frequency generating units shall have spurious and phase modulation components suppressed by at least 40 dB below rated output. Compatible operation shall be possible using not only the frequency standard group OA-4792/SRC23(V), but the AN/URQ-9 and AN/URQ-10 frequency standards as well.

3.2.5 Audio characteristics. Each transmitter input and receiver output audio signal connection shall be made to each sideband by balanced lines with 600-ohm source and input impedances.

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3.2.5.1 Audio switching. Audio switching shall be done at the Navy SSR and SRT switchboards.

3.2.5.2 Supervisory voltages. In addition to the voice and data normally carried by the wire pairs representing independent sideband transmitter inputs and receiver outputs, externally generated supervisory voltages shall be impressed between each signal pair and ground suitable for measurement at the center tap of the input and output transformers of the transmitter and receiver respectively.

3.2.6 Automatic tuning. The AN/SRC-23A(V) shall be capable of automatic tuning and remote operation to reduce tuning time and to facilitate remote manual or processor control. Control and operational status information shall be primarily digital in nature. Complete control capability shall be present locally at the R-T group OA-8012A/SRC-23(V) by means of front panel controls, readily accessible and visible. Complete remote control capability shall also be possible through the use of the R-T remote control group OA-4791/SRC-23(V).

3.2.7 System certification. The capability shall be provided to permit rapid certification that the system is operational and usable at any desired time by means of special test signals, unit samples and analyzing circuits capable of indicating the systems operational status either when operating in the communication circuit or when in standby condition.

3.2.8 Compatibility. Compatibility with standard Navy terminal equipment for SSB voice and teletypewriter operation is required. The AN/SRC-23A(V) shall be capable of compatible operation with the NTDS Data Terminal Sets AN/SSQ-29, and AN/USQ-36(V).

3.2.8.1 Remote control units. Each radio channel shall be capable of compatible operation with up to four C-1138A/UR remote control units. The required compatibility shall be in conformance with the characteristics of Radio Set Control C-1138A/UR specified in NAVSHIPS 0967-191-5010. Provisions shall include a 12 VDC power supply, either side of which shall be capable of being grounded.

3.2.9 Transmit-receive switching. Transmit-receive switching shall be accomplished by a handset push-to-talk button for voice, by break-in keying with release time delay for CW operation, and by a keyline at the teletypewriter control point.

3.2.9.1 "A" link switching. Operation of the radio in the NTDS "A" Link shall be simplex with T/R switching controlled by the supervisory voltages generated by the Data Terminal. The time required for T/R switching with full transmitter output stabilized shall not be greater than 7 milliseconds. Provisions shall be made to operate the receiver as a monitor of the transmission when in the transmit condition.

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3.3 Equipment characteristics.

3.3.1 System composition. The AN/SRC-23A shall be composed of equipment in the form of the following basic building blocks in combinations and quantity as specified in the order:

(a) R-T Group OA-8012A/SRC-23(V) consisting of:

(1) Transmitter, Radio T-1004/SRC-23(V).
(2) Receiver, Radio R-1361/SRC-23A.
(3) Transmitter, control C-4785A/SRC-23(V).
(4) Receiver, control C-4784A/SRC-23(V).
(5) Cabinet, Electrical Equipment CY-4932/SRC-23(V).
(6) Selector, Frequency C-4783/SRC-23(V).
(7) Amplifier, RF AN-3790A/SRC-23(V).

(b) Amplifier, RF AN-3799/SRC-23(V).
(c) R-T Remote Control Group OA-4791(V)/SRC-23(V) consisting of:

(1) Control Indicator, Transmitter C-6704/SRC-23(V).
(2) Control Indicator, Receiver C-6705/SRC-23(V).
(3) Monitor, Audio Frequency ID-1370/SRC-23(V).

(d) Frequency Standard Group OA-4792/SRC-23(V) consisting of:

(1) Quantity 3-Frequency Standard O-1107/SRC-16.
(2) Comparator, Signal CM-270/SRC-16.
(3) Cabinet, Electrical Equipment CY-4029/SRC-23(V).

(e) Test Set - Monitor, Radio TS-2476/SRC-23(V).

3.3.2 R-T group. The R-T group OA-8012A/SRC-23(V) shall be housed in a cabinet not exceeding 72 inches (182.88 cm) in height, 14 inches (35.56 cm) in width and 30 inches (76.20 cm) in depth.

3.3.2.1 Cooling. The R-T shall be water cooled.

3.3.2.2 Time meter. One elapsed time meter shall be provided as part of the equipment cabinet.

3.3.2.3 Transmitter. The transmitter T-1004/SRC-23(V) shall perform audio-to-RF translation and linear amplification of voice of multitone type input signals.

3.3.2.3.1 Transmitter design. The transmitter shall be designed so that there shall be substantially no disturbance to other transmitters during on-off and tuning periods when operating with the associated coupling networks of the AN/SRA-34(V).

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3.3.2.3.2 Transmitter tune-up. The transmitter and the associated coupling network of the AN/SRA-34(V) shall be capable of tune-up without actual operation on the radiating antenna with an emission of less than 50 microwatts on a nominal 50-ohm antenna.

3.3.2.3.3 Audio levels. The transmitter shall accept a voice input of minus 10 dBm average talker power input from the SRT switchboard which shall be raised to a 0 dBm voice signal level (based on a 0 dBm test tone) by means of a compressor amplifier.

3.3.2.3.3.1 Audio input signal. The audio signal input from the NTDS Data Terminal will be a complex waveform consisting of phase shifted tones in the 500 to 3000 Hz range spaced 110 Hz apart applied to either or both sidebands. The nominal level of the composite signal measured at the transmitter input shall be 0 dBm per sideband or plus 3 dBm for one sideband only. The composite signal shall have a peak-to-root-mean-square ratio of 10.3 dB, the value of which shall not be exceeded more than 0.1 percent of the time.

3.3.2.3.3.2 FSK mode binary keying. The transmitter shall be capable of being keyed in the FSK mode of operation by the binary information input from the teletypewriter equipment. The input signal will be on two lines with either line capable of being grounded at the teletypewriter equipment. The FSK output signal shall be a space for 0 ma-DC current in the input line and mark for 60 ma-DC current in the input line. Provision shall be made to optionally reverse the output mark and space frequencies corresponding to current conditions on the input keying line.

3.3.2.3.4 Phase stability. The average transmit RF or receive audio phase deviation over a 13.33 millisecond (ms) or 22 ms period shall not exceed 6 degrees between any two successive periods.

3.3.2.3.4.1 Absolute time delay. The absolute time delay for any signal within the range of 800 to 3050 Hz shall be between 0.7 and 1.2 ms. In the frequency range between 495 to 800 Hz, the time delay shall be between 0.7 and 1.6 ms.

3.3.2.3.5 Frequency response. The 2.5 dB passband shall be F_c +415 Hz to F_c +3050 Hz for the upper sideband and F_c -415 Hz to F_c -3050 Hz for the lower sideband. F_c is the dial or carrier frequency, and the response shall be that obtained at the antenna due to excitation at the audio inputs.

3.3.2.3.6 Output noise.

3.3.2.3.6.1 On frequency noise-on keyed with signals. At rated power output, hum and noise shall be at least 50 dB below peak envelope power (PEP) when measured in a 3 kHz bandwidth either side of the suppressed carrier frequency.

3.3.2.3.6.2 Off frequency noise-on keyed with signals. Up to and including rated power output using a two-tone test signal input, the maximum broadband noise as measured with a 3 kHz bandwidth receiver shall be 10 microvolts (μv) at any

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frequency between 2 and 30 MHz with the exception of those frequencies between ±15 percent of the operating frequency, harmonics of the operating frequency, and the HF oscillator frequency.

3.3.2.3.6.3 Off frequency noise-on keyed without signal. With the audio signal inputs terminated in 600-ohm dummy loads and the transmitter in the keyed condition maximum broadband noise as measured with 3 kHz bandwidth receiver shall be 2 μv at any frequency between 2 and 30 MHz excluding ±15 percent of the operating frequency and the HF oscillator frequency. Within 15 percent but excluding plus or minus 3 kHz of the operating frequency and under the above conditions the maximum broadband noise shall be 10 uv.

3.3.2.3.6.4 Off keyed noise. With the transmitter in the off keyed condition, the maximum broadband noise as measured with a 3 kHz bandwidth receiver shall be 1 uv at any frequency between 2 and 30 MHz.

3.3.2.3.6.5 Output noise measurement method. The characteristics specified in 3.3.2.3.6.4 shall be determined by measurements employing a signal substitution method wherein the receiver associated with RF equipment shall be used as the measurement receiver.

3.3.2.3.7 Gain. A CW tone of 7.1 dBm into either sideband shall drive the transmitter to rated average power output. The gains of each sideband shall be equal to within a tolerance of ±0.5 dB.

3.3.2.3.7.1 Gain reserve. The transmitter shall be designed to have a 6 dB reserve of gain over that which is required to maintain nominal output levels so that the transmitter can be compensated for losses of gain due to aging and other causes.

3.3.2.3.7.2 RF power control. The RF output power shall be continuously adjustable below rated power by means of a manual control in all modes except data.

3.3.2.3.8 Automatic load control (ALC). The ALC feedback shall be derived from the RF amplifier to prevent the RF amplifier being driven into intermodulation distortion greater than that specified. The ALC time constants shall be between 1 and 2 ms attack with approximately critical damping, and between 100 and 150 ms decay. The ALC loop shall maintain rated PEP output within ±1.0 dB as the input signal is increased from a level resulting in attainment of ALC threshold to a level of 6 dB higher.

3.3.2.3.9 Transmitter. The transmitter T-1004/SRC-23(V) shall be capable of generating an upper and a lower sideband in the frequency range of 2.0000 to 29.9999 MHz in integer increments of 100 Hz.

3.3.2.3.9.1 Impedance. The output load impedance shall be a nominal 50 ohms unbalanced. The input impedance, for each sideband, from 300 to 3050 Hz shall be 600 ohms (±20 percent) balanced with an ungrounded center tap and with a 0 to 15 ma DC balanced within ±10 percent flowing through each half of the primary

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winding in such a direction to be flux cancelling. The maximum DC resistance shall be 30 ohms each side of center.

3.3.2.3.9.2 Output power. When operating into a nominal 50-ohm load the rated output power shall be 0.4 watt (PEP) and shall not exceed a maximum power output of 4.0 watts (PEP).

3.3.2.3.9.3 Unwanted emission. The unwanted RF emission characteristics shall be in accordance with the following:

(a) Carrier suppression - at least 45 dB below rated PEP.
(b) Opposite sideband rejection - at least 50 dB below rated PEP.
(c) Harmonic suppression - at least 50 dB below rated PEP.
(d) Spurious suppression - at least 40 dB below rated PEP.
(e) Distortion - at least 35 dB below either tone of a standard two tone test signal.

3.3.2.3.9.4 Translator unit. The translator unit shall maintain its required performance while operating into the maximum standing-wave-ratio (SWR) characteristics of the AM-3790A/SRC-23(V) and the AM-3799/SRC-23(V) amplifier input circuits.

3.3.2.3.10 Power amplifier AM-3790A/SRC-23(V). The amplifier shall be capable of delivering 850 watts peak envelope and 425 watts average power continuously through the frequency range of 2.0000 to 29.9999 MHz.

3.3.2.3.10.1 Maximum SWR. The maximum SWR presented by the nominal 50-ohms unbalanced input impedance to the power amplifier shall be 1.4 to 1. The output impedance shall be 50 ohms unbalanced (coaxial) and capable of operation into a load of up to 1.3 to 1 SWR.

3.3.2.3.10.2 Distortion products. The amplifier, when operating from minimum to rated output power, shall not generate third or higher order distortion products at a level exceeding 30 dB below either tone of a standard two tone test signal.

3.3.2.3.10.3 Harmonic emission suppression. Second harmonic emission shall be suppressed by at least 35 dB and all other harmonic and spurious emissions shall be suppressed by at least 45 dB as measured at the RF terminals with the rated PEP as a reference.

3.3.2.4 Receiver, Radio R-1361/SRC-23(V). The receiver shall perform RF to audio frequency translation and linear amplification of voice or multitone type input signals.

3.3.2.4.1 Receiver. The receiver shall be a dual, SSB equipment with independent audio outputs from each or both the upper and lower sidebands within the frequency range of 2.0000 to 29.9999 MHz in integer 100 Hz increments with optional continuous tuning between the 100 Hz steps on all modes except data.

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3.3.2.4.2 Input. The input impedance shall be nominally 50 ohms.

3.3.2.4.2.1 Receiver gain control. A three-position receiver gain control shall be provided. One position shall insert a 100-ohm resistance in the receiver gain control line. The second position shall add an additional 40 dB pad. In the third position, both inserts shall be removed.

3.3.2.4.2.2 Receiver gain control. An automatic device shall be included to insure that inadvertent transmission by any equipment tuned to the receiver frequency cannot damage the receiver.

3.3.2.4.3 Output. The output for each sideband shall be transformer coupled, with an ungrounded center tap, balanced for audio frequencies in the range 300 to 3050 Hz at an impedance of 600 ohms (±10 percent). Maximum DC resistance of each side of the center tap shall be 30 ohms and the output shall be capable of carrying 30 ma of direct current at center tap with balanced current (unbalance not to exceed 10 percent in each half).

3.3.2.4.3.1 RMS audio output level. The RMS audio output level to the SSR switchboard shall be 0 dBm ±3 dB. The maximum peak-to-rms voltage ratio shall be 16 dB.

3.3.2.4.3.2 Receiver output hum and noise. The receiver output hum and noise, measured by a two-tone test from 5 uv to 0.5 volts rms, per tone, shall be 20 dB below either tone for a 55 cycle band.

3.3.2.4.3.3 Beat frequency oscillator. FSK signal receipt shall utilize an adjustable beat frequency oscillator (BFO) for compatible operation with the AN/URA-17B teletypewriter converter. The BFO shall also adjust the CW audio output frequency. The BFO stability shall be 0.1 percent.

3.3.2.4.4 Automatic gain control (AGC).

3.3.2.4.4.1 Time constants. The receiver AGC shall have an attack time constant of 6 to 12 ms, for a change from no signal to a two-tone 2.0 hard PEV 16 tone signal, to the level attained at a 20 hard μv (PEV) two-tone signal, shall be 10 to 20 ms. The AGC circuit shall be capable of repeating this operation every ms, provided the period between data signals is more than 10 ms.

3.3.2.4.4.2 Dynamic range. For antenna signal levels between 10 uv rms to 1 volt rms, the receiver AGC shall maintain the receiver output level at 0 dBm ±3 dB. Receiver AGC control voltage shall be developed separately for each sideband IF amplifier and the gain of each IF amplifier shall be controlled separately by these respective voltages. The AGC control voltage developed for each sideband IF amplifier shall be gated separately to control the gain of the RF amplifier so that the AGC voltage of greater magnitude controls. Of the 100 dB range of the AGC voltage, approximately 40 dB and 60 dB shall be linearly proportioned to the RF and IF stages respectively.

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3.3.2.4.5 Phase stability. The average transmit RF or receiver audio phase deviation over a 13.33 or 22 ms period shall not exceed 6 degrees between any two successive periods.

3.3.2.4.5.1 Absolute time delay. The absolute time delay for any signal within the range of 800 to 3050 Hz shall be between 0.7 and 1.2 ms. In the frequency range from 495 to 800 Hz, the time delay shall be between 0.7 and 1.6 ms.

3.3.2.4.6 Frequency response. The 2.5 dB passband of each sideband shall be 415 to 3050 Hz at the receiver output. The gain of each sideband shall be adjustable to within ±0.5 of the nominal value.

3.3.2.4.7 Sensitivity. The sideband CW sensitivity shall be 1 uv rms for a 10 dB signal plus noise-to-noise ratio (2750 Hz band).

3.3.2.4.8 Noise figure. The receiver noise figure shall be 17 dB or better.

3.3.2.4.9 Linearity. From the receiver input to the audio output, the peak of any intermodulation or harmonic components shall be 35 dB or more below the peak of either tone of a standard two-tone test signal for a per-tone signal of 50,000 uv rms input to the receiver. The above products shall be 14 dB or more below either tone, for a per-tone signal of 250,000 uv rms input.

3.3.2.4.10 Cross modulation. The cross modulation product shall be at least 30 dB below the output level of a desired single-tone signal of 200 uv rms, when an interfering two-tone signal of 3.0 volts rms per-tone is applied ±15 percent or more from the suppressed carrier frequency.

3.3.2.4.11 Desensitization. With a desired single-tone signal of 10 uv rms input the receiver output shall decrease not more than 3 dB when an undesired signal-tone of 6 volts rms is applied ±15 percent from the desired signal frequency. Broadband noise or undesired signal shall be less than 10 uv when measured in a 3 kHz band.

3.3.2.4.12 Spurious responses. The IF rejection shall be at least 100 dB for 500 kHz IF, with all other IF frequencies at least 90 dB. The image rejection shall be at least 80 dB for frequencies below 25 MHz and at least 65 dB for frequencies above 25 MHz. All other spurious responses shall be at least 60 dB down.

3.3.2.4.13 Squelch. Voice squelch characteristics compatible with SSB transmission shall be provided.

3.3.2.4.14 Simulated doppler shifts. Circuitry to produce continuously variable simulated doppler shifts of up to ±50 Hz shall be provided. Calibration accuracy shall be at least plus or minus 2 cycles and frequency stability shall be that of the BFD.

3.3.3 Power amplifier AM-3799/SRC-23(V). The requirements for this amplifier shall be identical to those of the power amplifier AM-3790A/SRC-23(V) except for the following:

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3.3.3.1 Average power. The amplifier shall be capable of delivering 4.25 KW PEP and 2125 watts average power continuously when connected to the transmitter T-1004/SRC-23(V) with 100 feet (30.48 m) maximum of RG-142B/U cable conforming to MIL-C-17.

3.3.3.2 Output impedance. The output impedance shall be 50 ohms unbalanced (coaxial) suitable for 2 to 1 SWR 50-ohm load. The input impedance shall be 50 ohms unbalanced coaxial with a maximum SWR of 1.5 to 1.

3.3.3.3 Distortion products. The third and higher order distortion products shall not exceed 35 dB below either tone of a standard two-tone test signal when the unit is operating from minimum to rated output power.

3.3.3.4 Emission suppression. Harmonic and spurious emissions shall be suppressed at least 50 dB as measured at the RF terminals with the rated PEP as a reference.

3.3.3.5 Amplifier housing. The amplifier shall be housed in a cabinet not exceeding 72 inches (182.88 cm) in height, 25 inches (63.5 cm) in width and 28 inches (71.12 cm) in depth.

3.3.3.6 Time meter. One elapsed time meter shall be provided as part of the equipment cabinet.

3.3.3.7 Switching. The AM-3799/SRC-23(V) power amplifier shall have the necessary circuitry to permit switchable operation with either, but not both, of two OA-8012A/SRC-23(V) T-R groups.

3.3.4 R-T remote control group OA-4791(V)/SRC-23(V). The control group shall consist of the equipment required for the remote control of the following functions of the transmitter and receiver from distances up to 900 feet (274.32 m):

(a) Transmitter frequency selection.
(b) Receiver frequency selection.
(c) Transmitter operating mode selection.
(d) Receiver operating mode selection.
(e) T-R audio level monitoring.
(f) Transmitter RF level monitoring.

3.3.4.1 Size. The control group shall not exceed a volume of 3 cubic feet and shall be configured for completely compatible mounting in the CY-4516/S cabinet specified in MIL-C-24056. Requirements specified for the equipment herein shall be met while mounted in and as an integral part of the CY-4516/S cabinet.

3.3.4.1.1 VWSR alarm. The R-T remote control group OA-4791(V)/SRC-23(V) shall incorporate a voltage-standing-wave-ratio (VSWR) alarm at the transmitter control indicator C-6704/SRC-23(V).

3.3.4.2 Tuning control. Frequency selection shall be by means of digital frequency information utilizing a 2-out-of-5 binary-coded decimal form for all decimal digits except the 100 cycle digits where straight binary coded decimal shall be used.

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3.3.4.2.1 Frequency selection. Frequency selection in 100 Hz increments shall be achieved by means of a 25-wire, time-shared bus (plus one ground wire and one tuning motor power wire, for digital tune enable, plus 4 wires for 100 cycle decimal digit per channel) for any number of channels.

3.3.4.2.2 Tune sequence control. Tune sequence control shall be achieved by means of diplexed supervisory and monitoring signals on the coaxial cable carrying the RF signals.

3.3.4.2.3 Tune retention. The controlled equipment shall retain its last tuned condition when the digital frequency information is removed and will not retune when the same information is reapplied.

3.3.4.2.4 Interlock. An interlock shall be provided to prevent simultaneous use of the time shared tuning bus.

3.3.4.2.5 Tune-up time. Tune-up time per transmit or receive channel (including the associated antenna couplers of AN/SRA-34A(V) shall be nominally 20 seconds and shall not exceed 60 seconds.

3.3.4.2.6 Other equipment. Provisions shall be made in the remote control units and in the controlled equipment to allow for incorporation of other remote control equipment and other radio channels respectively.

3.3.4.3 Transmitter control functions and indications. The transmitter control equipment in addition to tuning, shall provide facilities for the following control functions and indications:

(a) Mode selection.
(b) Radio silence tune selection.
(c) Amplifier selection.
(d) Transmitter gain.
(e) Standby-operate control.
(f) Local or remote control indication.
(g) Standby-tune-operate indication.
(h) Alarm indicators.

3.3.4.4 Receiver control functions and indicators. The receiver control equipment in addition to tuning, shall provide facilities for the following control functions and indications:

(a) Mode selection.
(b) BFO control.
(c) Receiver gain control.
(d) Squelch control.
(e) Standby-operate control.
(f) Local or remote indication.
(g) Standby-tune-operate indication.
(h) Alarm indicators.
(i) Simplex - duplex selection.

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3.3.4.5 Audio RF monitor. The audio RF monitor equipment shall provide for visual monitoring of the audio voltage on the transmit and receiver audio lines (4 pair), and forward and reflect RF power.

3.3.4.6 Local-remote selection. The local or remote control option selection shall be made at the controlled equipment.

3.3.5 Frequency standard group OA-4792/SRC-23(V). The frequency standard group shall consist of three standard signal generators plus a signal comparator. Two generators shall be in standby condition continuously and the output of any generator shall be capable of selection for reference use, for all AN/SRC-23(V) radio equipments and the NTDS terminal. The comparator shall be capable of comparing the output among the three generators or with an external standard.

3.3.5.1 Signal generator. The standard signal generator shall be capable of continuous unattended operation with output frequencies at 1 MHz and 100 kHz.

3.3.5.1.1 Signal output. The signal output at each of the frequencies specified in 3.3.5.1 shall be a minimum of one volt into a 50-ohm resistive load.

3.3.5.1.2 Crystal resonator. The resonator employed shall be a high precision quartz 5 MHz crystal.

3.3.5.1.3 Frequency stability. The stability characteristics of the generator shall be as follows:

(a) Aging rate: Less than one pp 10^8 for 60 days after 21-day warmup period.
(b) Temperature coefficient: Less than ±12 pp 10^{11} per degree C.
(c) Line voltage coefficient: Less than 1 pp 10^{11} per percent line voltage change.
(d) Reset accuracy: Better than 5 pp 10^{11}.
(e) Frequency change with shock: Less than 1 pp 10^8.
(f) Short-term stability: Average phase shift over any 22 millisecond period shall not exceed 6.0° for any two successive 22 ms periods, or 13.33 ms periods.

3.3.5.1.4 Harmonic distortion. Harmonic distortion shall be not greater than one percent. Other spurious signals shall be 78 dB below the specified output signal levels.

3.3.5.1.5 Frequency dividers. Frequency dividers shall be employed to reduce the resonator frequency to the required output frequencies. The design shall be so that output from the dividers will not be present unless the input is energized by the oscillator. Any interruption in input to the dividers shall cancel the output signal immediately and completely. A method for restarting the dividers shall be provided.

3.3.5.1.6 Internal standby power source. An internal standby power source shall supply power to the unit at all times during which other suitable power

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is not available. Standby power shall be automatically connected to the signal generator upon interruption of the primary power source without imparing the signal stability.

3.3.5.1.6.1 Standby power source battery. The standby power source shall consist of sealed nickel-cadmium batteries of sufficient capacity to allow the signal generator to operate continuously for two hours without any degradation in performance. During operation from the primary power source, the batteries shall be trickle charged.

3.3.5.1.7 Coarse frequency control. A coarse frequency control shall be provided capable of allowing variation of frequency over a range of ±1 pp 10^6.

3.3.5.1.8 Fine frequency control. A fine frequency control shall be provided capable of allowing variation of frequency in incremental steps of 1 pp 10^{10} for a total change of 500 pp 10^{10}.

3.3.5.2 Signal comparator. The signal comparator shall amplify and distribute the 100 kHz signal from any one of the three signal generators and provide a visual display of the frequency error between any two of the three signal generators or between any one of the three generators and an external source of standard frequency. Provisions shall be made for the following:

(a) Alarm circuits to indicate failure of vital functions.
(b) Automatic switching to maintain a 100 kHz signal on line at all times.
(c) Bypass of the comparator by manual operation of a switch.

3.3.5.2.1 Coaxial inputs. Four coaxial 50-ohm unbalanced inputs shall be provided. The input level will be 1.0 volt at 100 kHz.

3.3.5.2.2 Coaxial outputs. Coaxial 50-ohm unbalanced outputs shall be provided. Each output level shall be 1.0 volt at 100 kHz. Each of the outputs shall provide at least 40 dB isolation from the input.

3.3.5.2.3 Functions. The functions of the comparator shall in no way degrade the quality of the applied signals except spurious output level shall be at least 75 dB below the output level specified in 3.3.5.2.2.

3.3.5.2.4 Accuracy. The comparator shall be capable of comparing frequencies of the input specified in 3.3.5.2.3 to within 10^9 accuracy during not more than a 100 second sampling period.

3.3.5.3 Frequency standard group housing. The frequency standard group equipment shall be housed in a cabinet suitable for shelf mounting. The cabinet shall not exceed 10 inches (25.40 cm) in height, 24 inches (60.96 cm) in width and 24 inches (60.96 cm) in depth.

3.3.5.4 Malfunction alarm. The frequency standard group OA-4792/SRC-23(V) shall provide the capability to cause an alarm at the remote transmitter control unit and local transmitter control unit in the event that any one of two

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3.3.5.2 Frequency dividers. Frequency dividers shall be employed to

frequency standards, selected for operation through the frequency comparator unit, should have a malfunction or, if a malfunction is detected in the comparator unit which would create an alarm condition. Additionally, a frequency sensor for the 100 kHz input to the OA-8012A/SRC-23(V) shall be constantly sampled and, in the event of a loss of the 100 kHz input, cause an alarm condition at the local and remote transmitter control units.

3.3.6 Test set monitor, radio TS-2476/SRC-23(V). The Radio Test Set Monitor shall be capable of analyzing frequency distortion, performing system level checks, and performing frequency lock tests on both upper and lower sidebands of a communications channel. It shall also be capable of providing visual indication of FSK reception.

3.3.6.1 Size. The radio test set monitor unit shall not exceed a volume of 4 cubic feet and shall be configured for completely compatible mounting in the electrical equipment cabinet CY-4516/S specified in MIL-C-24056.

3.3.6.1.1 Integral mounting. The unit shall conform to applicable detail requirements of this specification while mounted and as an integral part of the CY-4516/S cabinet.

3.3.6.2 Input and output switching. Input and output switching shall be done at the shipboard SRR and SRT switchboards.

3.3.6.3 Test set. The test set portion of the radio test-set monitor shall provide one tone of two tones for the LSB and USB during the test-tone R-R frequency-locking check, one tone of two tones for the LSB, USB or both sidebands during system level checks, and two tones on either LSB or USB to measure the level below a tone of the 400 Hz, 800 Hz, F1 + F2 intermodulation product, third order (2F1 - F2, 2F2 - F.) intermodulation products, and second harmonic of the F1 tone.

3.3.6.3.1 Test set input. The input characteristics shall be as follows:

(a) Frequency range: 300 to 3050 Hz.
(b) Input impedance: 600 ohms ±20 percent balanced for both inputs.
(c) Level: Nominal, 0.707 ±0.2 vrms. Maximum, 1.5 volts rms.

3.3.6.3.2 Test set output. The output characteristics shall be as follows:

(a) Frequencies: 1275 and 1550 Hz, + 0.2 percent.
(b) Impedance: 600 ohms ±20 percent balanced over 1000 to 2000 Hz range.
(c) The signal amplitude shall be stable to within 1 percent of reference signal.
(d) Hum and noise shall be 70 dB below a one tone output of 0.775 volts rms or a two tone output of 0.55 volts rms per tone.
(e) Distortion Mode of operation:

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(1) Level: 2 ±0.2 volts rms maximum per tone of two tone signal.
(2) Distortion: Harmonics and intermodulation products shall be at least 70 dB below a one tone output of 0.775 volts rms or a two tone output of 0.55 volts rms per tone.

(f) Level check mode of operation:

(1) Level: Nominal, 2 ±0.1 volts rms per tone of two tone signal; Maximum, 3 ±0.5 volts rms for a single tone signal.
(2) Distortion: Harmonics and intermodulation products shall be at least 60 dB below a one tone output of 3 volts rms per tone output of 2 volts rms per tone.

3.3.6.3.3 Test set controls and indicators. The following controls and indicators shall be provided:

(a) Individual level controls for each audio output and a two section step attenuator (0 to 100 dB attentuation in 10 dB attenuation in 1 dB steps) for controlling the level of both audio outputs simultaneously.
(b) One control to adjust the audio input level to the audio frequency filters and a two-section step attenuator (0 to 40 dB attenuation in 10 dB steps plus 0 to 10 dB attenuation in 1 dB steps) for controlling the level of the audio input to the VU amplifier.
(c) An internal screwdriver adjustment for attenuation compensation of each audio frequency filter without changing the filter input and output impedance.
(d) An "A" scale (0 to minus 20 VU and 0 to plus 3 VU) VU meter to monitor signal levels. The meter shall be protected from prolonged inputs or more than +7 dBm signals.

3.3.6.3.4 Test set monitor. The test-set monitor unit shall be capable of measuring the hum distortion signal levels as follows:

Hum and distortion signal level below a tone	Degree of accuracy
50 dB or less	±1 dB
60 dB	-6 dB
70 dB	-10 dB

3.3.6.4 Monitor scope. A cathode ray oscilloscope shall be provided for the purpose of displaying received test tones and FSK signals in order to monitor and adjust FSK operating audio frequencies and to perform RF lock checks on each radio channel.

3.3.6.4.1 Visual display. A visual display of lissajous patterns of at least 1.5 inches (3.81 cm) shall be possible when input levels of the audio signals are 0 dBm ±3 dB from a balanced 600-ohm source.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at any time. Verification will include (a) surveillance of the operations to determine that practices, methods, and procedures are being properly applied, (b) Government product inspection to measure quality or product to be offered for acceptance, and (c) Government product inspection of delivered products to assure compliance with all inspection requirements of this specification. Failure of the contractor to promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been made or until conformance of product to prescribed criteria has been demonstrated.

4.1.2 Quality assurance terms and definitions. Quality assurance terms used in this specification shall be as defined in MIL-STD-109.

4.2 Classification of inspection. The method of examination and testing shall be classified as follows:

(a) First Article Inspection (4.3)
(b) Quality Conformance Inspection (4.4)

(1) Production Inspection (4.4.1)
(2) Environmental Inspection (4.4.2)

(c) Reliability Testing (4.5)
(d) Inspection of Preparation for Delivery (4.3)

4.3 First article inspection. First article inspection shall consist of all examination and testing necessary to determine conformance with the requirements of this specification. First article inspection shall include the examination and testing specified in production inspection, and environmental inspection.

4.4 Quality conformance inspection.

4.4.1 Production inspection. Production inspection shall be made on every equipment offered for delivery. The inspection shall comprise such examination and testing as will prove the workmanship and reveal the omissions and errors of the production process such as functional and performance tests at a limited number of points, tests which detect deviations from design, tests of adjustment,

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and tests which detect hidden defects of material. Production inspection shall include the requirements of paragraph 3.2 and 3.3.

4.4.2 Environmental inspection. Environmental inspection shall be as listed and shall be performed on units which have been subjected to and passed production inspection.

Table I. Environmental inspection.

Inspection	Reference paragraph
Low temperature	3.1.1
High temperature	3.1.1
Steady state voltage and frequency	3.1.7
Transient voltage	3.1.7
Transient frequency	3.1.7
Spike voltage	3.1.7
Power interruption	3.1.7

4.4.2.1 Sampling for environmental inspection. One equipment shall be selected from each successive 50 equipments produced or fraction thereof. The first sample equipment shall be selected from the first months production.

4.4.3 Inspection of preparation for delivery. Inspections shall be conducted to insure conformance with the requirements of Section 5 of this specification.

4.5 Reliability test. The equipment shall be subjected to a 700 hour reliability test in accordance with test plan XXVIII of MIL-STD-781. The test shall be in accordance with test level C of MIL-STD-781 except that the off cycle shall be one hour per day. Input voltage cycling shall be required.

4.5.1 Test sample. The total operating test time shall be 700 hours to be accumulated on one unit or divided between two or more units. The test shall be completed prior to the delivery of production equipment.

4.5.2 Failure. Failures are any malfunction or parameter deviation which prevents the equipment from performing within the operational requirements specified herein.

4.5.3 Evaluation of test. No specific accept/reject criteria is established. The contractor shall evaluate each observed failure to determine the cause. Where equipment manufacturer design or part manufacturer's design deficiencies are identified or when pattern failures (as defined by MIL-STD-781) exist, the contractor shall propose and implement corrective action to prevent or reduce the probability of recurrence. The test shall be considered successfully completed when all corrective actions have been approved by the procuring activity.

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5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced document listed in Section 2 do not apply unless specifically stated in the contract or order. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order.)

5.1 Preservation, packaging, packing and marking shall be in accordance with MIL-E-17555.

6. NOTES

6.1 Intended use. This equipment is intended for shipboard use.

6.2 Environmental inspection. Approval to ship may be withheld at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action.

6.2.1 Initial shipments. Initial shipment of end items will be contingent upon successful completion of the environmental inspection.

Preparing activity:
Navy - EC

(Project 5820-N657)

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   A. GIVE PARAGRAPH NUMBER AND WORDING.

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